

Chapter 4: Visual Elements and Graphics

Horizontal Rule Element

Configures a horizontal line

`<hr>`



Figure 4.1 The `<hr>` tag configures a horizontal line

CSS border Property

Configures a border on the top, right, bottom, and left sides of an element

Consists of

- border-width
- border-style
- border-color

h2 { border: 2px solid #ff0000 }

Heading with Border

CSS Borders: Block / Inline Elements

- Block display element
 - Default width of element content extends to browser margin (or specified width)
- Inline display element
 - Border closely outlines the element content

```
h2 { border: 2px solid #ff0000; }
```

```
a { border: 2px solid #ff0000; }
```

Heading with Border

[This anchor tag has a border](#)

Border Display Can Vary by Browser

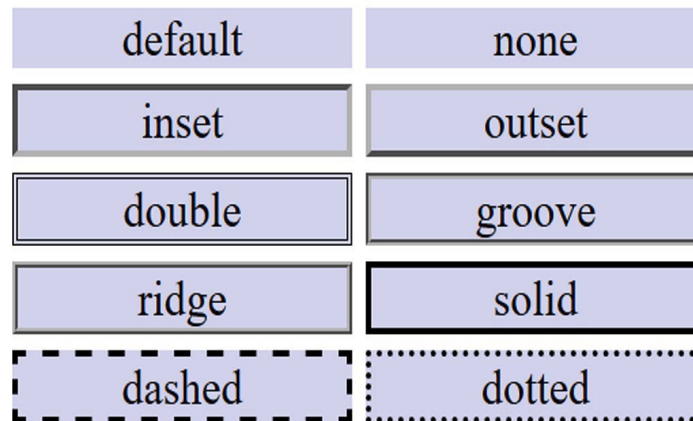


Figure 4.2 Examples of the various border-style values rendered by Firefox

Configuring Specific Sides of a Border

Use CSS to configure a line on one or more sides of an element

- border-bottom
- border-left
- border-right
- border-top

```
h2 { border-bottom: 2px solid #ff0000 }
```

Heading with Border

CSS padding Property

Configures empty space between the content of the HTML element and the border

Set to 0px by default

```
h2 { border: 2px solid #ff0000; padding: 5px; }
```

Heading with Border

No padding property configured:

Heading with Border

Configure Padding on Specific Sides of an Element

Use CSS to configure padding on one or more sides of an element

- padding-bottom
- padding-left
- padding-right
- padding-top

```
h2 { border: 2px solid #ff0000; background-color: #cccccc; padding-left: 5px;  
padding-bottom: 10px; padding-top: 10px; }
```



Heading with Border

CSS padding Property Shorthand: two values

Two numeric values or percentages

- first value configures top and bottom padding
- the second value configures left and right padding

```
h2 { border: 2px solid #ff0000; background-color: #cccccc; padding: 20px 10px;}
```



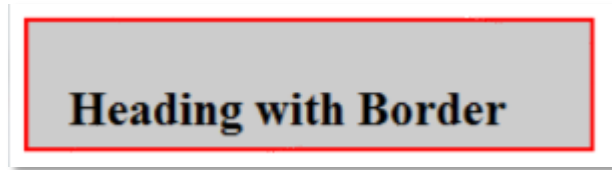
Heading with Border

CSS padding Property Shorthand: four values

Four numeric values or percentages

- Configure top, right, bottom, and left padding

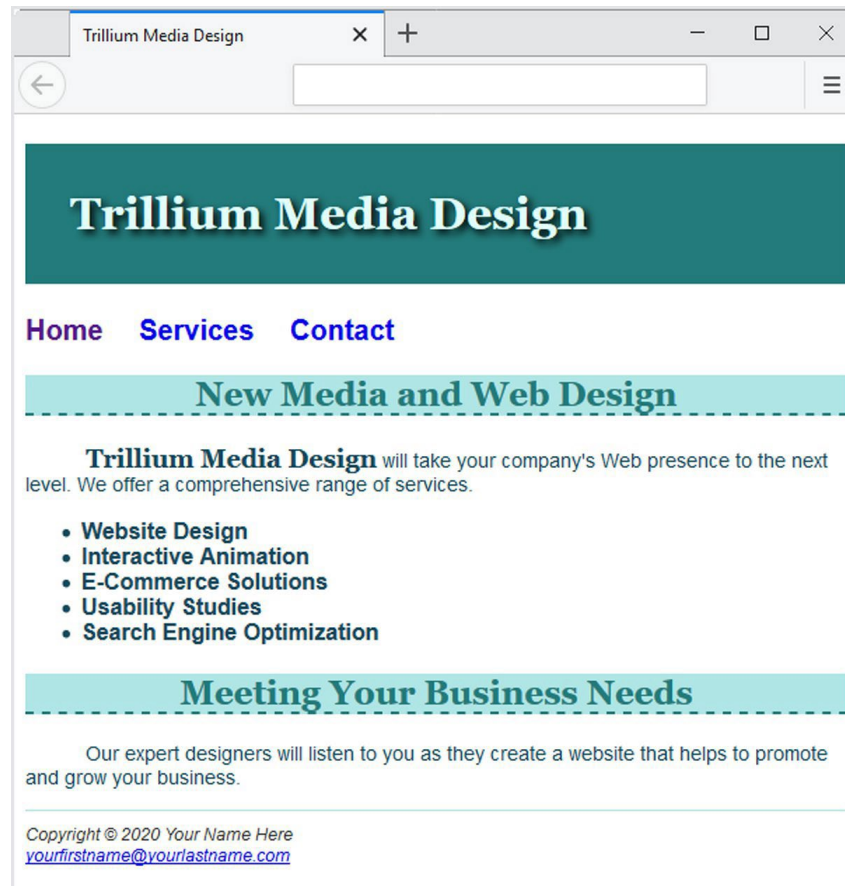
```
h2 { border: 2px solid #ff0000; width: 250px; background-color: #cccccc; padding: 30px 10px 5px 20px; }
```



Hands-On Practice

```
h2 { background-color: #B0E6E6; color: #237B7B;  
      font-family: Georgia, "Times New Roman", serif; text-align: center;  
      border-bottom: 2px dashed #237B7B; }
```

Figure 4.3 CSS border and padding properties add visual interest to the page.

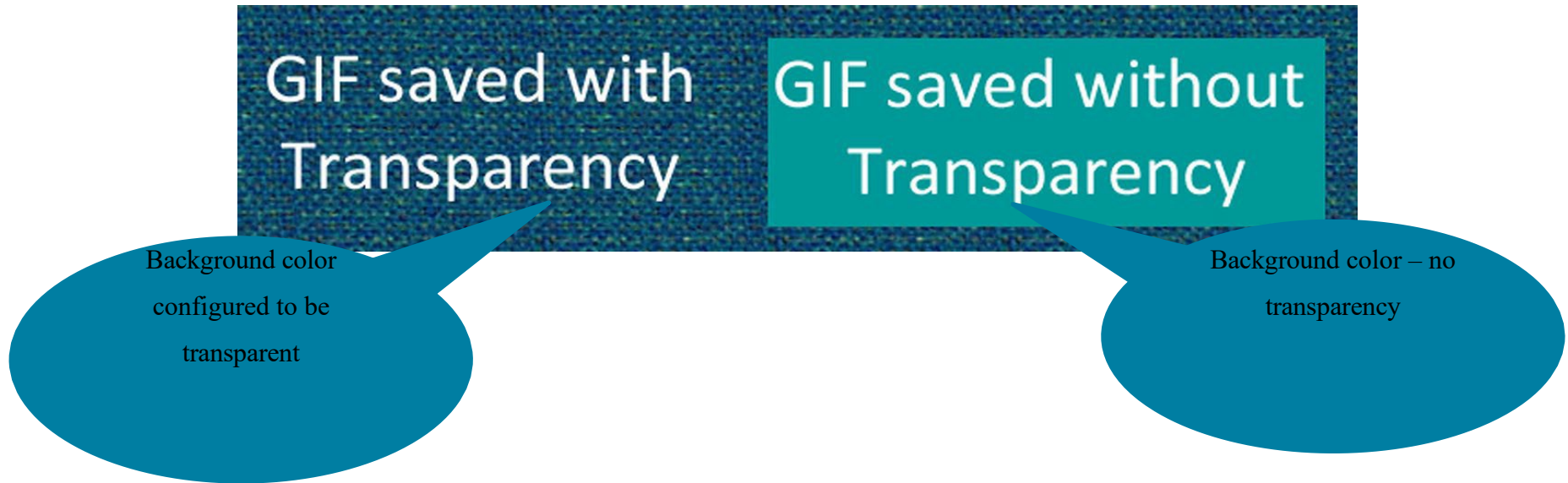


Types of Graphics

Graphic types commonly used on web pages:

- GIF
- JPG
- PNG
- WebP

Figure 4.6 Comparison of transparent and nontransparent GIFs



Types of Graphics

- Graphics Interchange Format
- Best used for line art and logos
- Maximum of 256 colors
- One color can be configured as transparent
- Can be animated
- Uses lossless compression
- Can be interlaced

JPEG

- Joint Photographic Experts Group
- Best used for photographs
- Up to 16.7 million colors
- Use lossy compression
- Cannot be animated
- Cannot be made transparent
- Progressive JPEG – similar to interlaced display

Figure 4.7 A JPEG saved at 80% quality (55KB file size) displays well on a web page



PNG

- Portable Network Graphic
- Supports millions of colors
- Supports multiple levels of transparency
(but most browsers do not – so limit to one transparent color for Web display)
- Supports interlacing
- Uses lossless compression
- Combines the best of GIF & JPEG
- Well-supported by modern browsers

WebP

- <https://developers.google.com/speed/webp/>
- Supports millions of colors
- Supports transparency
- Uses both lossy and lossless compression
- Intended to provide improved compression for GIF and JPG images
- Browser support is growing

HTML Image Element

- Configures graphics on a web page

- src Attribute
 - File name of the graphic
- alt Attribute
 - Configures alternate text description
- height Attribute
 - Height of the graphic in pixels
- width Attribute
 - Width of the graphic in pixels



Figure 1.3 Who really updated that web page you are viewing?

Accessibility & Images

Required:

- Configure the alt attribute
 - Alternate text content to convey the meaning/intent of the image
 - NOT the file name of the image
 - Use alt="" for purely decorative images

Recommended:

- If your site navigation uses image links for the main navigation, provide simple text links at the bottom of the page.

Image Link

- To create an image hyperlink use an anchor element to contain an image element



```
<a href="index.html"></a>
```

- Some browsers automatically add a border to image links.
- Configure CSS to eliminate the border
img { border-style: none; }

Thumbnail Image

A small image configured to link to a larger version of that image.

```
<a href="island.jpg"></a>
```

Figure 4.7 A JPEG saved at 80% quality (55KB file size) displays well on a web page

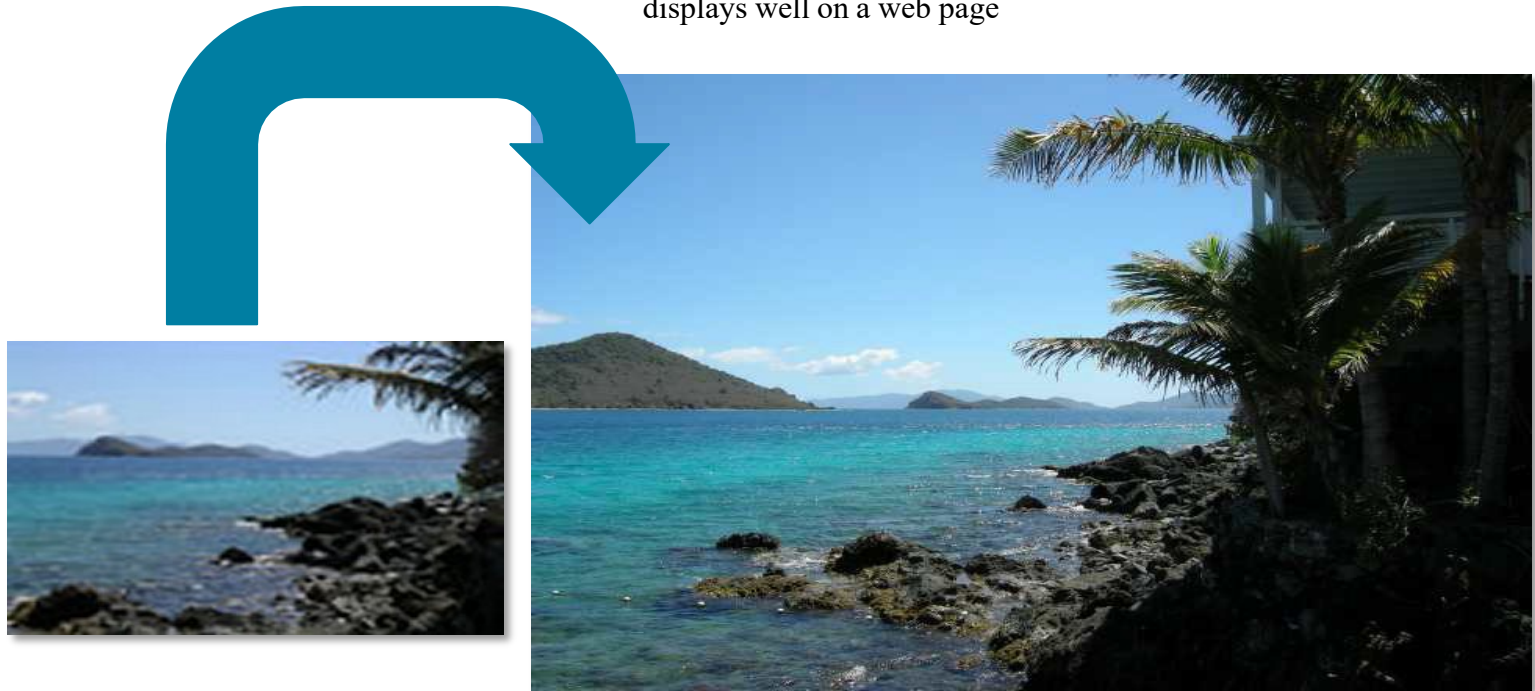


Figure 4.9 This small thumbnail image is only 5KB

Image Optimization

The process of creating an image with the lowest file size that still renders a good quality image—**balancing image quality and file size**.

Photographs taken with digital cameras are not usually optimized for the Web

Optimize An Image for the Web

Image Optimization

- Reduce the file size of the image
- Reduce the dimensions of the image to the actual width and height of the image on the web page.

Image Editing Tools:

- GIMP <https://www.gimp.org/> (FREE!)
- Adobe Photoshop
- Pixlr
 - <https://pixlr.com/x/> (FREE!)

Choosing Names for Image Files

Use all lowercase letters

Do not use punctuation symbols and spaces Do not change the file extensions

(should be .gif, .jpg, .jpeg, or .png)

Keep your file names short but descriptive

- i1.gif is probably too short
- myimagewithmydogonmybirthday.gif is too long
- dogbday.gif may be just about right

Organizing Your Site

- Place images in their own folder
- Code the path to the file in the src attribute

```

```

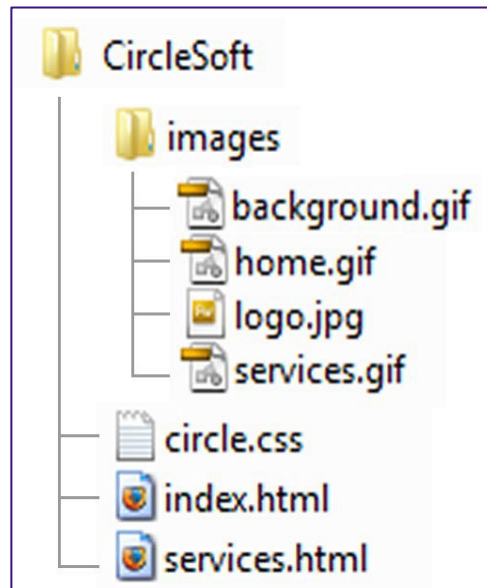


Figure 4.21 A folder named “images” contains the graphic files

CSS background-image Property

Configures a background-image

By default, background images tile (repeat)

```
body { background-image: url(background1.gif); }
```

Figure 4.17 A long, thin background image tiles down the page

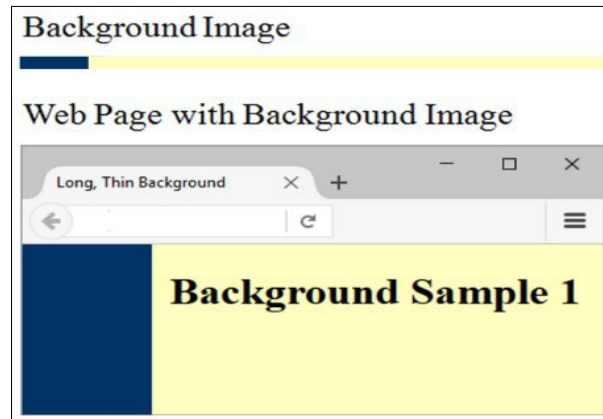
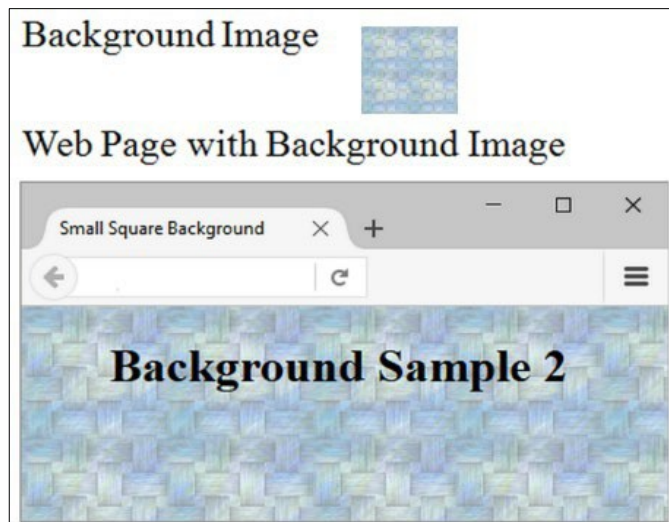


Figure 4.18 A small rectangular background is repeated to fill the web page window



CSS background-repeat Property

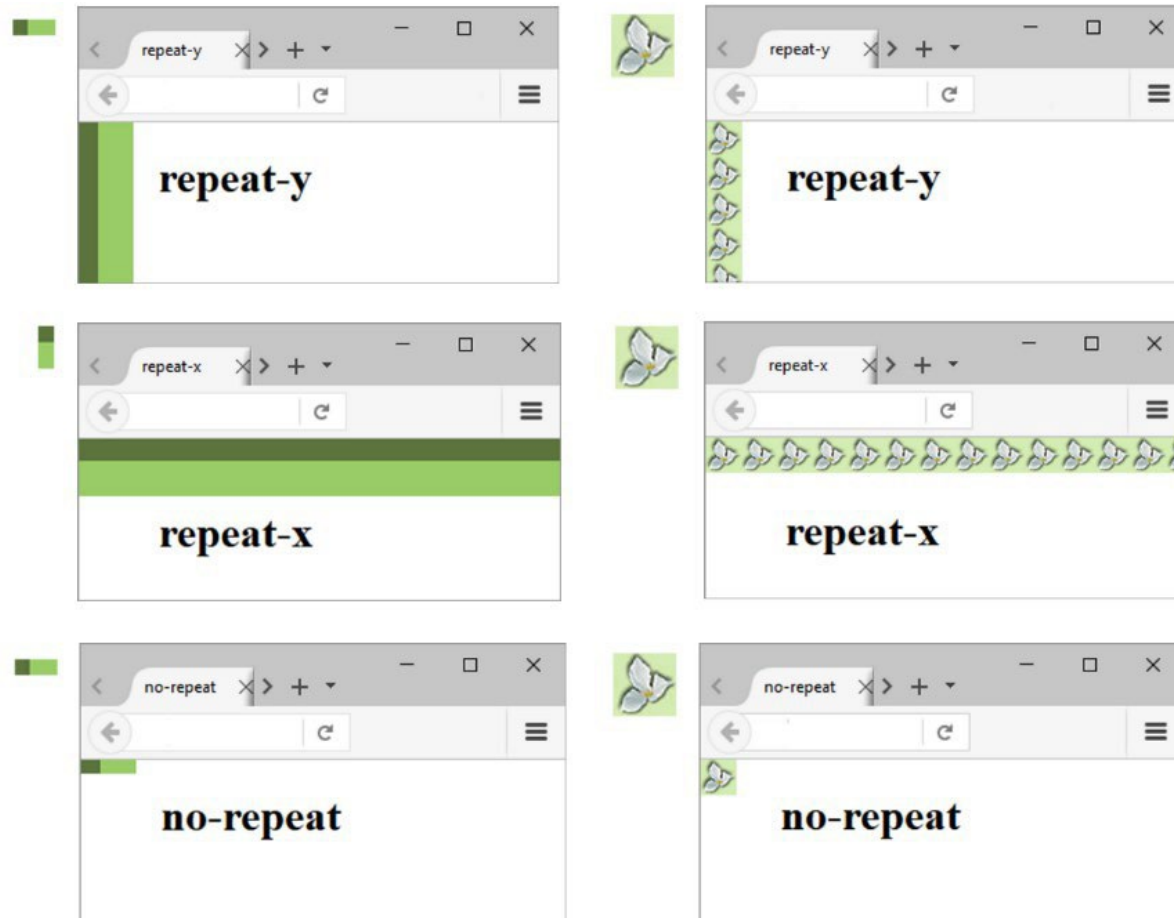


Figure 4.19 Examples of the CSS background-repeat property

Using background-repeat

flower.gif:



```
h2 { background-color: #d5edb3; color: #5c743d;  
      font-family: Georgia, "Times New Roman", serif; padding-left: 30px;  
      background-image: url(flower.gif); background-repeat: no-repeat;  
    }
```

CSS background-position Property

The default placement of a background image is top left.

The background-position property can change this placement.



```
h2 { background-image: url(flower.gif); background-position: right; background-repeat: no-repeat; }
```

CSS background-clip Property

- confines the display of the background image

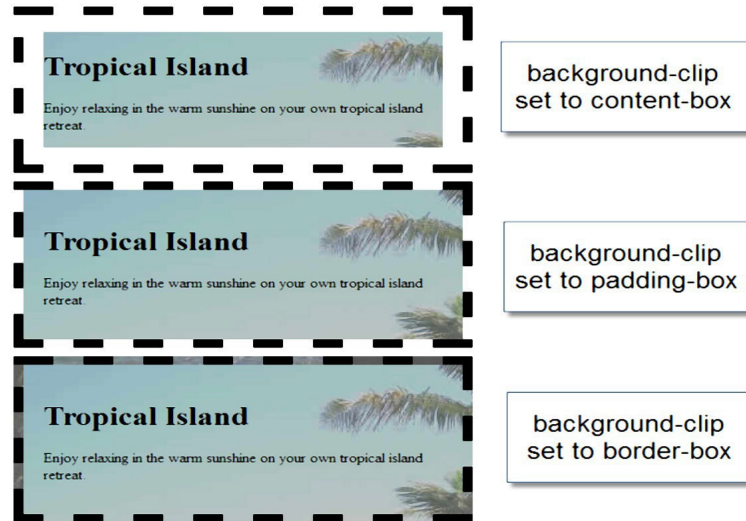


Figure 4.23 The CSS background-clip property

CSS background-origin Property

- positions the background image relative to the content, padding or border

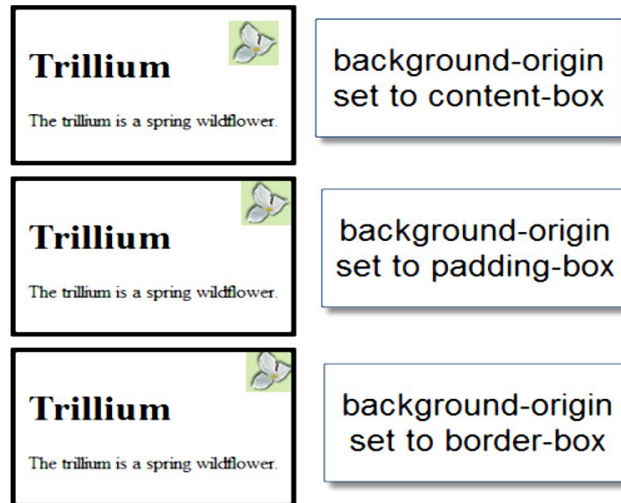
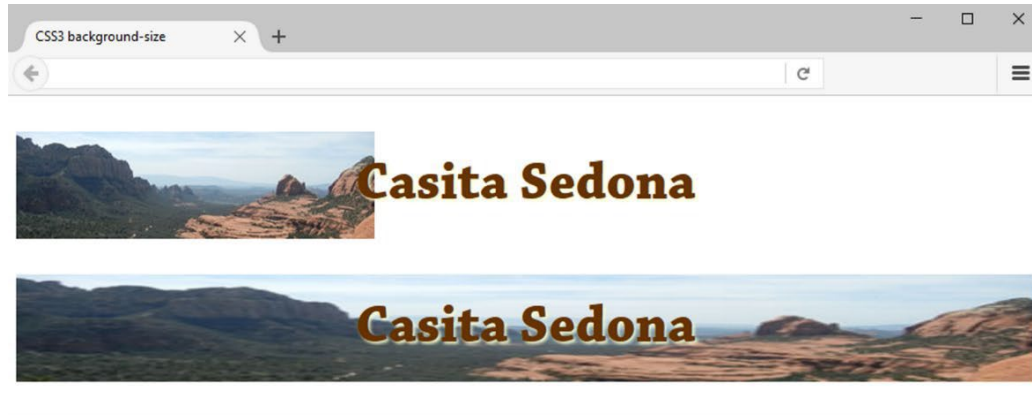


Figure 4.24 The CSS background-origin property

CSS background-size Property

can be used to resize or scale the background image



Without
background-size property

background-size: 100% 100%;

Figure 4.25 Configuring a background image.

CSS Multiple Background Images

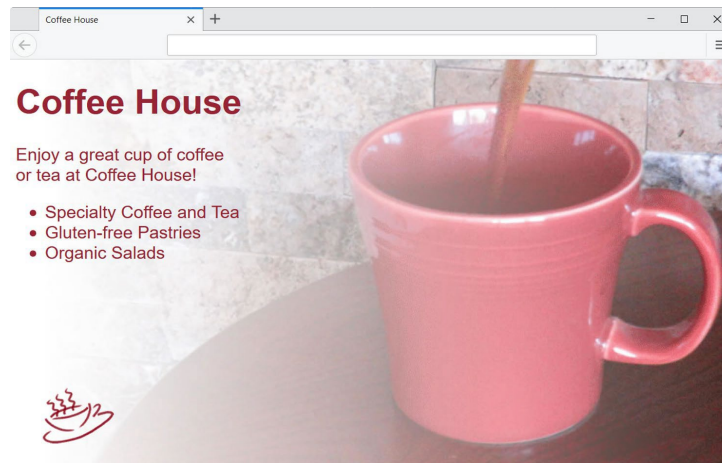
Use the background property to configure multiple background images.

Example:

```
body { background: url(coffee.gif) no-repeat left bottom,
```

```
fixed;                                url(coffeepour.jpg) no-repeat background-size: auto, cover; }
```

Figure 4.27 The browser displays multiple background images.



CSS Rounded Corners

border-radius property

- Configures the horizontal radius and vertical radius of the corner
- Numeric value(s) with unit (pixel or em) or percentage

Example

```
h1 { border-radius: 15px;}
```

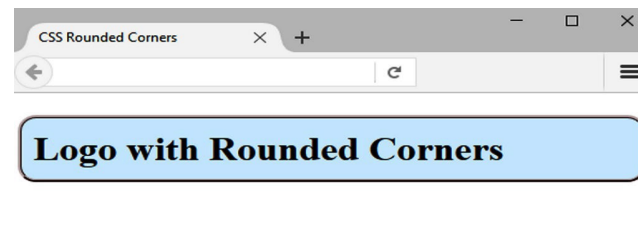


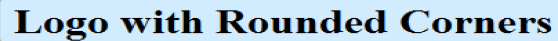
Figure 4.32 Rounded corners were configured with CSS

Examples of Rounded Corners

One value for border-radius configures all four corners

Example:

```
border-radius: 15px;
```

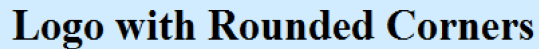


Logo with Rounded Corners

Four values for border-radius configure each corner separately Ordered by top left, top right, bottom right, bottom left

Example:

```
border-radius: 15px 30px 100px 5px;
```



Logo with Rounded Corners

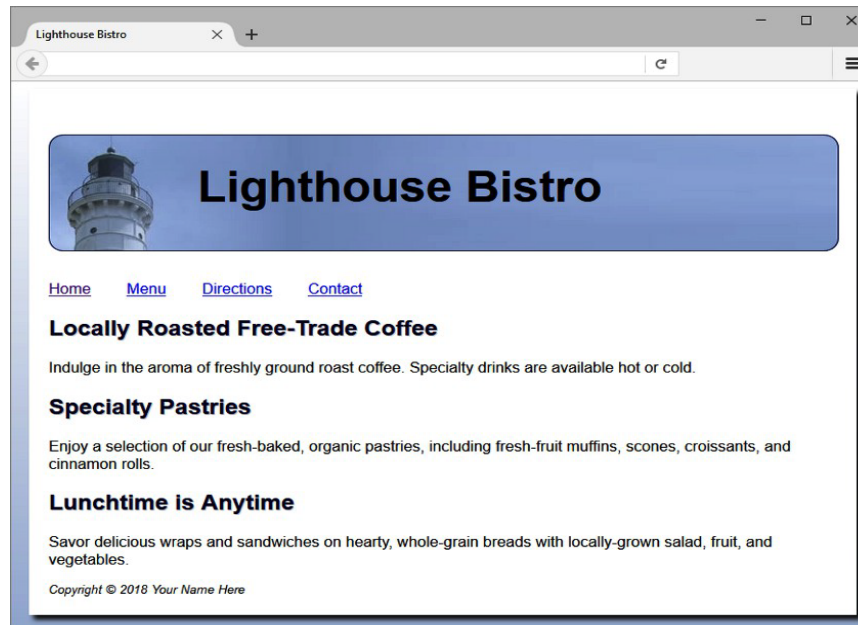
CSS box-shadow Property

Configure the horizontal offset, vertical offset, blur radius, and valid color value

Example:

```
#wrapper { box-shadow: 5px 5px 5px #828282; } Optional keyword: inset
```

Figure 4.35 The centered neutral background along with shadow properties add dimension



CSS opacity Property

Configure the transparency of the element Opacity range:

- 0 Completely Transparent
- 1 Completely Opaque

horizontal offset, vertical offset, blur radius, and valid color value

Example:

```
h1 { background-color: #FFFFFF; opacity: 0.6; }
```

Figure 4.37 The background of the h1 area is transparent

